

Theoretical Physics 2 – Sheet 08

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19. Time evolution and measurements of a spin with $s = \frac{1}{2}$ We consider a spin- $\frac{1}{2}$ particle with initial state

$$|\psi(0)\rangle = |\downarrow\rangle.$$

The Hamiltonian is

$$\hat{H} = \frac{\hbar\omega}{2}\sigma_x.$$

- (a) At time $t_1 = \frac{\pi}{2\omega}$ the operator σ_z is measured. In which state is the particle at this time? Which measurement results are possible, and with which probabilities do they occur?

Calculation

The time evolution operator is obtained from the Schrödinger equation by

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar}\hat{H}t\right) = \exp\left(-i\frac{\omega t}{2}\sigma_x\right).$$

Since $\sigma_x^2 = \mathbb{I}$, the exponential splits into even and odd powers.

Calculation

$$\begin{aligned}\hat{U}(t) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-i\frac{\omega t}{2}\sigma_x\right)^n \\ &= \mathbb{I} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \left(\frac{\omega t}{2}\right)^{2k} - i\sigma_x \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} \left(\frac{\omega t}{2}\right)^{2k+1} \\ &= \mathbb{I} \cos\left(\frac{\omega t}{2}\right) - i\sigma_x \sin\left(\frac{\omega t}{2}\right).\end{aligned}$$

At $t_1 = \frac{\pi}{2\omega}$ this gives $\frac{\omega t_1}{2} = \frac{\pi}{4}$, hence

Calculation

$$\begin{aligned}|\psi(t_1)\rangle &= \left[\mathbb{I} \cos\left(\frac{\pi}{4}\right) - i\sigma_x \sin\left(\frac{\pi}{4}\right)\right] |\downarrow\rangle \\ &= \frac{1}{\sqrt{2}} (|\downarrow\rangle - i|\uparrow\rangle).\end{aligned}$$

The eigenstates of σ_z are

$$\sigma_z |\uparrow\rangle = +|\uparrow\rangle, \quad \sigma_z |\downarrow\rangle = -|\downarrow\rangle.$$

Therefore the two possible measurement results are $+1$ and -1 .

Calculation

$$P(+1) = |\langle \uparrow | \psi(t_1) \rangle|^2 = \left| -\frac{i}{\sqrt{2}} \right|^2 = \frac{1}{2},$$

$$P(-1) = |\langle \downarrow | \psi(t_1) \rangle|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

Answer

Immediately before the measurement,

$$|\psi(t_1)\rangle = \frac{1}{\sqrt{2}} (|\downarrow\rangle - i|\uparrow\rangle).$$

A measurement of σ_z gives

$$+1 \text{ with probability } \frac{1}{2}, \quad -1 \text{ with probability } \frac{1}{2}.$$

After the measurement the state collapses to $|\uparrow\rangle$ if the result was $+1$, and to $|\downarrow\rangle$ if the result was -1 .

- (b) Directly after the measurement the system evolves for a time interval $\Delta t = \frac{\pi}{2\omega}$. At time $t_2 = t_1 + \Delta t = \frac{\pi}{\omega}$, σ_z is measured again. Depending on the result from part (a), in which state is the particle at t_2 , and with which probabilities do the possible results occur?

Calculation

After the first measurement, there are two possible collapsed states. For both branches the further time evolution is given by

$$\hat{U}(\Delta t) = \mathbb{K} \cos\left(\frac{\omega\Delta t}{2}\right) - i\sigma_x \sin\left(\frac{\omega\Delta t}{2}\right) = \frac{1}{\sqrt{2}}(\mathbb{K} - i\sigma_x),$$

because $\Delta t = \frac{\pi}{2\omega}$.

Calculation

If the first result was $+1$, then directly after the first measurement

$$|\psi'\rangle = |\uparrow\rangle.$$

Thus

$$\begin{aligned} |\psi(t_2)\rangle &= \hat{U}(\Delta t) |\uparrow\rangle \\ &= \frac{1}{\sqrt{2}} (\mathbb{K} - i\sigma_x) |\uparrow\rangle \\ &= \frac{1}{\sqrt{2}} (|\uparrow\rangle - i|\downarrow\rangle). \end{aligned}$$

Therefore

$$P(+1 \mid +1 \text{ at } t_1) = \frac{1}{2}, \quad P(-1 \mid +1 \text{ at } t_1) = \frac{1}{2}.$$

Calculation

If the first result was -1 , then directly after the first measurement

$$|\psi'\rangle = |\downarrow\rangle.$$

Thus

$$\begin{aligned} |\psi(t_2)\rangle &= \hat{U}(\Delta t) |\downarrow\rangle \\ &= \frac{1}{\sqrt{2}} (\mathbb{K} - i\sigma_x) |\downarrow\rangle \\ &= \frac{1}{\sqrt{2}} (|\downarrow\rangle - i|\uparrow\rangle). \end{aligned}$$

Therefore

$$P(+1 \mid -1 \text{ at } t_1) = \frac{1}{2}, \quad P(-1 \mid -1 \text{ at } t_1) = \frac{1}{2}.$$

Answer

In both branches the second measurement has equal probabilities:

$$P(+1) = \frac{1}{2}, \quad P(-1) = \frac{1}{2}$$

conditional on the respective state after the first measurement. The difference between the two branches is the relative phase and which basis vector is written first.

(c) In contrast to the previous parts, the particle now evolves without interruption until t_2 . In

which state is the particle at this time? Which probabilities result from a measurement of σ_z at t_2 ? Explain the difference to part (b).

Calculation

Without an intermediate measurement, the state is evolved directly from $t = 0$ to $t_2 = \frac{\pi}{\omega}$. Then

$$\frac{\omega t_2}{2} = \frac{\pi}{2}.$$

Hence

Calculation

$$\begin{aligned} |\psi(t_2)\rangle &= \left[\mathbb{1} \cos\left(\frac{\pi}{2}\right) - i\sigma_x \sin\left(\frac{\pi}{2}\right) \right] |\downarrow\rangle \\ &= -i\sigma_x |\downarrow\rangle \\ &= -i|\uparrow\rangle. \end{aligned}$$

The global phase $-i$ has no physical effect. Therefore this state is physically the same as $|\uparrow\rangle$.

Answer

At t_2 the state is

$$|\psi(t_2)\rangle = -i|\uparrow\rangle.$$

Therefore a measurement of σ_z gives

$$P(+1) = 1, \quad P(-1) = 0.$$

The difference to part (b) is caused by the intermediate measurement. In part (b), the first measurement destroys the coherent superposition and projects the state onto one of the σ_z eigenstates. Without this projection, the amplitudes continue to interfere coherently, and the final state becomes $|\uparrow\rangle$ up to a global phase.

20. System with angular momentum $l = 1$ We consider a system with angular momentum $l = 1$. The basis vectors

$$|+1\rangle, \quad |0\rangle, \quad |-1\rangle$$

are eigenvectors of $\hat{L}_z$ with eigenvalues $\hbar, 0, -\hbar$. The ladder operators satisfy

$$\hat{L}_{\pm} |m\rangle = \hbar\sqrt{2} |m \pm 1\rangle, \quad \hat{L}_+ |1\rangle = \hat{L}_- |-1\rangle = 0.$$

The Hamiltonian is

$$\hat{H} = \frac{\omega_0}{\hbar} (\hat{L}_u^2 - \hat{L}_v^2),$$

where $\hat{L}_u$ and $\hat{L}_v$ are the components along the diagonals of the xz -plane.

- (a) Give the matrix representation of $\hat{H}$ in the basis $\{|+1\rangle, |0\rangle, |-1\rangle\}$. What are the stationary states and their energy eigenvalues? Write them as $|E_1\rangle, |E_2\rangle, |E_3\rangle$ with $E_1 \geq E_2 \geq E_3$.

Calculation

Since u and v are the diagonals of the xz -plane, we use

$$\hat{L}_u = \frac{1}{\sqrt{2}}(\hat{L}_x + \hat{L}_z), \quad \hat{L}_v = \frac{1}{\sqrt{2}}(\hat{L}_z - \hat{L}_x).$$

Because operators do not necessarily commute, we expand the squares carefully.

Calculation

$$\begin{aligned} \hat{L}_u^2 - \hat{L}_v^2 &= \frac{1}{2}(\hat{L}_x + \hat{L}_z)^2 - \frac{1}{2}(\hat{L}_z - \hat{L}_x)^2 \\ &= \frac{1}{2}(\hat{L}_x^2 + \hat{L}_x\hat{L}_z + \hat{L}_z\hat{L}_x + \hat{L}_z^2) \\ &\quad - \frac{1}{2}(\hat{L}_z^2 - \hat{L}_z\hat{L}_x - \hat{L}_x\hat{L}_z + \hat{L}_x^2) \\ &= \hat{L}_x\hat{L}_z + \hat{L}_z\hat{L}_x. \end{aligned}$$

In the ordered basis $(|+1\rangle, |0\rangle, |-1\rangle)$, the matrices are

Calculation

$$\hat{L}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad \hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-) = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Therefore

Calculation

$$\begin{aligned} \hat{H} &= \frac{\omega_0}{\hbar}(\hat{L}_x\hat{L}_z + \hat{L}_z\hat{L}_x) \\ &= \frac{\hbar\omega_0}{\sqrt{2}} \left[\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} \right] \\ &= \frac{\hbar\omega_0}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}. \end{aligned}$$

To diagonalise, write

$$\hat{H} = \hbar\omega_0 A, \quad A = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}.$$

The characteristic polynomial of A is

Calculation

$$\det(A - \lambda \mathbb{I}) = -\lambda(\lambda^2 - 1).$$

Thus

$$\lambda_1 = 1, \quad \lambda_2 = 0, \quad \lambda_3 = -1.$$

The normalized eigenvectors are

Calculation

$$\begin{aligned} |E_1\rangle &= \frac{1}{2} | +1 \rangle + \frac{1}{\sqrt{2}} | 0 \rangle - \frac{1}{2} | -1 \rangle, & E_1 &= \hbar\omega_0, \\ |E_2\rangle &= \frac{1}{\sqrt{2}} | +1 \rangle + \frac{1}{\sqrt{2}} | -1 \rangle, & E_2 &= 0, \\ |E_3\rangle &= \frac{1}{2} | +1 \rangle - \frac{1}{\sqrt{2}} | 0 \rangle - \frac{1}{2} | -1 \rangle, & E_3 &= -\hbar\omega_0. \end{aligned}$$

Answer

$$\hat{H} = \frac{\hbar\omega_0}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

with stationary states and energies

$$\begin{aligned} |E_1\rangle &= \frac{1}{2} | +1 \rangle + \frac{1}{\sqrt{2}} | 0 \rangle - \frac{1}{2} | -1 \rangle, & E_1 &= \hbar\omega_0, \\ |E_2\rangle &= \frac{1}{\sqrt{2}} (| +1 \rangle + | -1 \rangle), & E_2 &= 0, \\ |E_3\rangle &= \frac{1}{2} | +1 \rangle - \frac{1}{\sqrt{2}} | 0 \rangle - \frac{1}{2} | -1 \rangle, & E_3 &= -\hbar\omega_0. \end{aligned}$$

(b) At time $t = 0$ let the system be in the state

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}} (| +1 \rangle - | -1 \rangle).$$

What is the state vector $|\psi(t)\rangle$? At time t , $\hat{L}_z$ is measured. What are the probabilities for

the different possible results?

Calculation

First we express the initial state in the energy eigenbasis. Using the eigenvectors from part (a),

Calculation

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}} (|E_1\rangle + |E_3\rangle).$$

The time evolution in the energy eigenbasis is diagonal:

Calculation

$$\begin{aligned} |\psi(t)\rangle &= \frac{1}{\sqrt{2}} \left(e^{-iE_1 t/\hbar} |E_1\rangle + e^{-iE_3 t/\hbar} |E_3\rangle \right) \\ &= \frac{1}{\sqrt{2}} \left(e^{-i\omega_0 t} |E_1\rangle + e^{i\omega_0 t} |E_3\rangle \right). \end{aligned}$$

Let $\theta = \omega_0 t$. Transforming back to the $\hat{L}_z$ basis gives

Calculation

$$\begin{aligned} |\psi(t)\rangle &= \frac{\cos \theta}{\sqrt{2}} | +1 \rangle - i \sin \theta | 0 \rangle - \frac{\cos \theta}{\sqrt{2}} | -1 \rangle \\ &= \frac{\cos(\omega_0 t)}{\sqrt{2}} (| +1 \rangle - | -1 \rangle) - i \sin(\omega_0 t) | 0 \rangle. \end{aligned}$$

A measurement of $\hat{L}_z$ can give the eigenvalues $\hbar, 0, -\hbar$. The probabilities are the squared moduli of the corresponding coefficients.

Calculation

$$\begin{aligned} P(L_z = \hbar) &= \frac{1}{2} \cos^2(\omega_0 t), \\ P(L_z = 0) &= \sin^2(\omega_0 t), \\ P(L_z = -\hbar) &= \frac{1}{2} \cos^2(\omega_0 t). \end{aligned}$$

Answer

$$|\psi(t)\rangle = \frac{\cos(\omega_0 t)}{\sqrt{2}} (| +1 \rangle - | -1 \rangle) - i \sin(\omega_0 t) | 0 \rangle$$

and

$$P(\hbar) = \frac{1}{2} \cos^2(\omega_0 t), \quad P(0) = \sin^2(\omega_0 t), \quad P(-\hbar) = \frac{1}{2} \cos^2(\omega_0 t).$$

- (c) Calculate the expectation values $\langle \hat{L}_x(t) \rangle$, $\langle \hat{L}_y(t) \rangle$ and $\langle \hat{L}_z(t) \rangle$. Which motion does the vector $\langle \hat{L}(t) \rangle$ perform? Sketch the trajectory of $\langle \hat{L}(t) \rangle$.

Calculation

We use the state from part (b), written as a column vector in the ordered basis $(|+1\rangle, |0\rangle, |-1\rangle)$:

$$|\psi(t)\rangle = \begin{pmatrix} \frac{\cos \theta}{\sqrt{2}} \\ -i \sin \theta \\ -\frac{\cos \theta}{\sqrt{2}} \end{pmatrix}, \quad \theta = \omega_0 t.$$

The angular momentum matrices are

Calculation

$$\hat{L}_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \hat{L}_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \hat{L}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

For $\hat{L}_z$, the probabilities of $m = +1$ and $m = -1$ are equal, hence the contributions cancel:

Calculation

$$\langle \hat{L}_z(t) \rangle = \hbar \frac{\cos^2 \theta}{2} - \hbar \frac{\cos^2 \theta}{2} = 0.$$

For $\hat{L}_x$, the contributions from the $|+1\rangle \leftrightarrow |0\rangle$ and $|0\rangle \leftrightarrow |-1\rangle$ terms cancel because the coefficients of $|+1\rangle$ and $|-1\rangle$ have opposite signs:

Calculation

$$\langle \hat{L}_x(t) \rangle = 0.$$

For $\hat{L}_y$, we compute explicitly:

Calculation

$$\begin{aligned}\hat{L}_y |\psi(t)\rangle &= \frac{\hbar}{\sqrt{2}} \begin{pmatrix} -i(-i \sin \theta) \\ i \frac{\cos \theta}{\sqrt{2}} - i \left(-\frac{\cos \theta}{\sqrt{2}}\right) \\ i(-i \sin \theta) \end{pmatrix} \\ &= \begin{pmatrix} -\frac{\hbar \sin \theta}{\sqrt{2}} \\ i \hbar \cos \theta \\ \frac{\hbar \sin \theta}{\sqrt{2}} \end{pmatrix}.\end{aligned}$$

Hence

$$\begin{aligned}\langle \hat{L}_y(t) \rangle &= \begin{pmatrix} \frac{\cos \theta}{\sqrt{2}} & i \sin \theta & -\frac{\cos \theta}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} -\frac{\hbar \sin \theta}{\sqrt{2}} \\ i \hbar \cos \theta \\ \frac{\hbar \sin \theta}{\sqrt{2}} \end{pmatrix} \\ &= -\frac{\hbar}{2} \sin \theta \cos \theta - \hbar \sin \theta \cos \theta - \frac{\hbar}{2} \sin \theta \cos \theta \\ &= -2\hbar \sin \theta \cos \theta \\ &= -\hbar \sin(2\theta).\end{aligned}$$

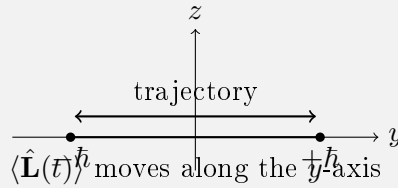
Answer

$$\langle \hat{L}_x(t) \rangle = 0, \quad \langle \hat{L}_y(t) \rangle = -\hbar \sin(2\omega_0 t), \quad \langle \hat{L}_z(t) \rangle = 0.$$

Therefore

$$\langle \hat{\mathbf{L}}(t) \rangle = (0, -\hbar \sin(2\omega_0 t), 0).$$

The expectation value vector oscillates on the y -axis between $-\hbar \mathbf{e}_y$ and $+\hbar \mathbf{e}_y$ with angular frequency $2\omega_0$.



- (d) At time t , a measurement of $\hat{L}_z^2$ is performed.
- Are there times at which only one single result is possible?
 - Assume that the measurement gives the result $\hbar^2$. In which state is the system immediately after the measurement? Describe the subsequent evolution without calculation.

Calculation

The operator $\hat{L}_z^2$ has eigenvalues

$$\hat{L}_z^2 | +1 \rangle = \hbar^2 | +1 \rangle, \quad \hat{L}_z^2 | 0 \rangle = 0, \quad \hat{L}_z^2 | -1 \rangle = \hbar^2 | -1 \rangle.$$

Thus the result $\hbar^2$ corresponds to the two-dimensional subspace spanned by $| +1 \rangle$ and $| -1 \rangle$, while the result 0 corresponds to the one-dimensional subspace spanned by $| 0 \rangle$.

From part (b),

$$|\psi(t)\rangle = \frac{\cos(\omega_0 t)}{\sqrt{2}}(| +1 \rangle - | -1 \rangle) - i \sin(\omega_0 t) | 0 \rangle.$$

Hence

Calculation

$$\begin{aligned} P(L_z^2 = \hbar^2) &= \cos^2(\omega_0 t), \\ P(L_z^2 = 0) &= \sin^2(\omega_0 t). \end{aligned}$$

Therefore only one result is possible when one of these probabilities is equal to 1.

Calculation

$$\begin{aligned} P(L_z^2 = \hbar^2) = 1 &\iff \cos^2(\omega_0 t) = 1 \iff \omega_0 t = n\pi, \\ &\iff t = \frac{n\pi}{\omega_0}, \quad n \in \mathbb{Z}, \\ P(L_z^2 = 0) = 1 &\iff \sin^2(\omega_0 t) = 1 \iff \omega_0 t = \frac{\pi}{2} + n\pi, \\ &\iff t = \frac{\pi}{2\omega_0} + \frac{n\pi}{\omega_0}, \quad n \in \mathbb{Z}. \end{aligned}$$

If the measurement gives $\hbar^2$, we project onto the subspace spanned by $| +1 \rangle$ and $| -1 \rangle$ and then normalize.

Calculation

$$\begin{aligned} \hat{P}_{\hbar^2} |\psi(t)\rangle &= \frac{\cos(\omega_0 t)}{\sqrt{2}}(| +1 \rangle - | -1 \rangle), \\ |\psi_{\text{after}}\rangle &= \frac{\hat{P}_{\hbar^2} |\psi(t)\rangle}{\text{norm } \hat{P}_{\hbar^2} |\psi(t)\rangle} = \frac{1}{\sqrt{2}}(| +1 \rangle - | -1 \rangle), \end{aligned}$$

up to an irrelevant overall phase. This is the same form as the initial state in part (b).

Answer

(i) Yes. For

$$t = \frac{n\pi}{\omega_0}$$

only the result $\hbar^2$ is possible. For

$$t = \frac{\pi}{2\omega_0} + \frac{n\pi}{\omega_0}$$

only the result 0 is possible.

(ii) If the result is $\hbar^2$, the state immediately after the measurement is

$$|\psi_{\text{after}}\rangle = \frac{1}{\sqrt{2}}(|+1\rangle - |-1\rangle)$$

up to a global phase. Since this is the same state as the initial state in part (b), the subsequent time evolution repeats the same oscillation: probability is periodically transferred between the $\hbar^2$ subspace and the 0 eigenstate $|0\rangle$.